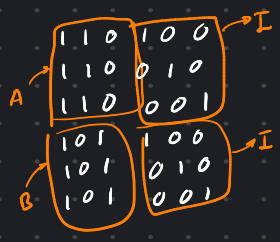
॥ श्रीगणेशाय नमः ॥

RBR 01

Consider the following row vectors: α_1= (1,1,0,1,0,0) α_2= (1,1,0,0,1,0) α_3= (1,1,0,0,0,1)

α_4= (1,0,1,1,0,0) α_5= (1,0,1,0,1,0) α_6= (1,0,1,0,0,1) The dimension of the vector space spanned by these row vectors is



$$\begin{bmatrix} A & I \\ B & I \end{bmatrix}$$

$$C = \begin{pmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{pmatrix}$$

Now let $A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix}$

So, $C = \begin{pmatrix} A & I \\ B & I \end{pmatrix} \to \begin{pmatrix} A & I \\ B-A & 0 \end{pmatrix}$

Here its clear that the rows in $(A \quad I)$ are Linearly Independent. So the rank of C will depend on the dimension of row space of $B - A$ which is equal to $\begin{pmatrix} 0 & -1 & 1 \\ 0 & -1 & 1 \\ 0 & -1 & 1 \end{pmatrix}$ Since all rows of this matrix is exactly the same so, the dimension of its row space=1. So , Required dimension=$3 + 1$.

Find a basis for Span(S) where

$$S = \left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ -2 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ 6 \\ -2 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix} \right\}.$$

a) ✓✓ $\left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 6 \\ -2 \end{bmatrix} \right\}$ b) $\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}$ c) $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$ d) $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$

$$\begin{bmatrix} 1 & -1 & 2 & 1 \\ 2 & -2 & 6 & 1 \\ 1 & -1 & -2 & 3 \end{bmatrix} \to \begin{bmatrix} 1 & -1 & 2 & 1 \\ 1 & -1 & 8 & -2 \\ 0 & 0 & -4 & 2 \end{bmatrix} \to \begin{bmatrix} 1 & -1 & 2 & 1 \\ 1 & -1 & 4 & 0 \\ 0 & 0 & 2 & -1 \end{bmatrix} \to \begin{bmatrix} 1 & -1 & 2 & 1 \\ 0 & 0 & 2 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

1st and 3rd colm are enough
1st and 4th colm are enough
(2nd, 3rd) or (2nd, 4th) are fine as well.

How many matrices are there in the vector space $M_{3\times4}(Z_2)$?

$\mathbb{Z}_2$ is a finite field consisting of two elements: 0 and 1. The operations of addition and multiplication are defined modulo 2.

- Addition: $x + y = (x + y) \mod 2$
- Multiplication: $x \cdot y = (x \cdot y) \mod 2$

For $M_{3\times4}(\mathbb{Z}_2)$, each entry in the matrix can be 0 or 1. There are 12 entries.
So # of matrices = 2^{12}.

4 The matrix

$$A = \begin{bmatrix} \frac{3}{2} & 0 & \frac{1}{2} \\ 0 & -1 & 0 \\ \frac{1}{2} & 0 & \frac{3}{2} \end{bmatrix}$$

has three distinct Eigen values and one of its Eigen vectors is

$$\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

. Which one of the following can be another Eigen vector of A?

The eigenvectors of a symmetric matrix A corresponding to different eigenvalues are orthogonal to each other.

The given matrix,

$$A = \begin{bmatrix} \frac{3}{2} & 0 & \frac{1}{2} \\ 0 & -1 & 0 \\ \frac{1}{2} & 0 & \frac{3}{2} \end{bmatrix}$$

$A^T = A$

Hence A is a symmetric matrix.

$$\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}^T = [1 \quad 0 \quad 1]$$

$$[1 \quad 0 \quad 1] \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = [1 + 0 - 1] = [0]$$

Therefore $\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$ can be another Eigenvector of A.